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1 Evaluation of management factors affecting the relative success of a Brook Trout eradication program
2 using YY male fish and electrofishing suppression

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Abstract

Removal of non-native fish populations can be crucial to the conservation of native species, but often presents a complex challenge for managers. The goal of Trojan Y chromosome (TYC) programs is to skew the non-native sex ratio until only males remain, leading to eradication. We present results from a simulation model used to explore effects of alternative management approaches on an in-progress mechanical removal and TYC program to eradicate non-native Brook Trout. Simulation results indicated that stocking fingerling YY males (~137 mm) was more effective than stocking catchable-sized YY males (~230 mm), although questions about inter-cohort competition warrant further investigation. Increasing the proportion of mature fingerling YY males reduced treatment time by increasing the number of YY male spawners and increasing density-dependent mortality on young, mature wild Brook Trout. Maximizing the spatial distribution of YY male releases may be crucial to program success but is also dependent upon immediate dispersal movements. Principles derived from our results can be broadly applied to the management of other aquatic invaded systems using TYC programs to eradicate non-native species.

Keywords: Conservation management; Invasive species biology; Simulation modeling; Stream fisheries; Trojan chromosomes;

33 Introduction

34 The establishment and spread of non-native fish species present significant challenges for the
35 conservation and management of natural resources (Havel et al. 2015). One approach to the removal of
36 non-native species is to manipulate the population sex ratio until only one sex remains, leading to
37 eventual eradication (Harvey-Samuel et al. 2017). Trojan Y chromosome (TYC) programs rely on the
38 reproductive contribution of introduced individuals with two Y chromosomes (i.e., YY males), resulting in
39 all-male offspring. With consistent releases over time, the number of females is expected to decline
40 through successive generations until the population consists solely of males. While the production of YY
41 males in aquaculture has been well-developed (Mair et al. 1997), their application as a fisheries
42 management tool has been relatively recent and lacks sufficient field testing (but see Kennedy et al.
43 2018). Given the resources required to implement a TYC program, tools that can be used to support
44 decision analysis for management actions are essential to developing a robust management strategy.

45 The use of simulation models to support the implementation of a TYC programs in an adaptive
46 management framework is a promising approach for optimizing strategies to achieve eradication of non-
47 native aquatic species. Such models can be a critical component of a successful adaptive management
48 program by informing initial management decisions, evaluating alternative hypotheses, and measuring
49 outcome uncertainty (Smith et al. 2013; Walters 1986). In such an application, model results can be
50 compared to observed empirical outcomes resulting from a management action in order to further
51 refine the model to accurately reflect current knowledge of underlying processes. The updated model is
52 then used to experimentally test management scenarios not yet implemented in the field, and to further
53 reduce uncertainty associated with alternative management decisions. Thus, the simulation model is an
54 integral component of the adaptive management feedback cycle (Bearlin et al. 2002).

Several earlier simulation models have demonstrated the feasibility of the TYC strategy for the eradication of non-native fish species, though few have been tied to a real-world system with ongoing TYC management. For example, Gutierrez and Teem (2006) used mathematical modeling (i.e., a system of differential equations) to demonstrate population eradication given constant releases of YY fish. Subsequent theoretical models explored the effects of factors such as the use of YY males versus YY females (Parshad et al. 2013), the dendritic configuration of the stream system (Gutierrez et al. 2012), fitness of the YY fish (Senior et al. 2013), and Allee effects (Beauregard et al. 2020). Day et al. (2020) investigated how factors related to the life history and ecology of YY fish affected the probability of eradicating a target fish population, but did not evaluate the effects of alternative management decisions on eradication outcomes.

In this paper we simulate alternative management actions in support of an adaptive management program designed to eradicate non-native Brook Trout (*Salvelinus fontinalis*) in a mountainous stream system where they occur in sympatry with native Westslope Cutthroat Trout (*Oncorhynchus clarkii lewisi*). Native to eastern North America, Brook Trout have been repeatedly introduced throughout western North America over the past 150 years, primarily stocked for recreational purposes (Fuller et al. 1999). Subsequent to stocking, the range of Brook Trout has extended beyond initial release locations (Paul and Post 2001), and the species is now widespread in western North America, often overlapping with and threatening the viability of native salmonids such as Cutthroat Trout and Bull Trout (*Salvelinus confluentus*; Dunham et al. 2002). As a result, Brook Trout are commonly targeted by management agencies for population suppression or eradication to facilitate the restoration of native salmonids (Buktenica et al. 2013; Shepard et al. 2002). Where Brook Trout and native salmonids are sympatric, the complete eradication of Brook Trout can be extremely difficult via mechanical removal alone (Meyer et al. 2006; Peterson et al. 2008). Accordingly, several state agencies are currently incorporating the use of YY males into their Brook Trout management plans. Given the

recency of this method, little is known about the optimal management application of YY males for reducing non-native populations while conserving native populations. Simulation models can provide vital support to such efforts as a vehicle for experimentation and analysis that cannot yet be accomplished in the field.

In this study, we present results from simulation modeling used to explore uncertainty associated with several management factors of a Brook Trout TYC program. Factors evaluated include number of YY males released, body size of YY males released, number of release locations, frequency of electrofishing suppression, and proportion of YY males mature at release. Specifically, our goal was to understand how varying each of these factors might affect the rate of population decline and time to eradication of the non-native species. While our model focuses on the management of Brook Trout, the simulation framework is flexible and can be adapted to any aquatic species where movement rates and demography can be described. Principles derived from our simulation results can likewise be broadly applied to the management of other invaded stream systems.

Materials and Methods

Study system

The model extent of this study included the occupied range of Brook Trout in the Sullivan Creek watershed, comprising 32.3 km of stream length. Sullivan Creek is a tributary to the Lower Pend Oreille River in northeastern Washington State, USA (Figure 1). Although non-native, Brook Trout were widely stocked throughout the Lower Pend Oreille Basin from 1902 until 1981 (Bowers 1904; Smith 1914; Kloempken 1996). Native Westslope Cutthroat Trout occur sympatrically with Brook Trout throughout much of their range in the Sullivan Creek watershed. However, Westslope Cutthroat Trout populations often extend further upstream than do Brook Trout populations due to the presence of barriers or high gradient reaches in most tributaries to Sullivan Creek that prevent upstream Brook Trout movement.

One of the fish management goals within the Sullivan Creek Basin is the suppression and removal of Brook Trout to aid restoration of Westslope Cutthroat Trout throughout the watershed. Electrofishing suppression of Brook Trout began in 2016, followed by the annual release of YY male Brook Trout in the headwaters of Sullivan Creek beginning in 2018 (Baker et al. 2020).

Model Description

To conduct simulations, we developed a system model for Brook Trout in Sullivan Creek using the spatially explicit, individual-based software CDMetaPOP (Cost Distance MetaPOPulation) version 1.44 (Landguth et al. 2017; Mims et al. 2019). For a complete description of prior applications of this Brook Trout model, as well as a description of incorporation of YY males into the model, see Day et al. (2018) and Day et al. (2020), respectively. Brook Trout parameters associated with demography (e.g., fecundity, growth) and behavior (e.g., dispersal, mating) were derived from empirical data. For a full list of parameter values and sources included in the model, as well as information on age and size structure of the simulated Brook Trout population, see Supplementary Material. Following, we provide only a brief description of the main processes in the YY male Brook Trout system model.

We divided Sullivan Creek and its tributaries into 151 model stream reaches (i.e., patches) covering the range of Brook Trout in the watershed (Figure 1), which were then populated with individual fish. Patches averaged 200 m in stream length and corresponded to existing sampling units delineated for the monitoring and management of fish populations in Sullivan Creek. Patches were characterized by spatially explicit factors including carrying capacity, stream temperature, spawning habitat, and suppression schedule. We modeled physical barriers (i.e., dams, culverts) that limited upstream movement based on the estimated permeability of the barrier. All movement (i.e., dispersal, mating) was based on a draw from a probability distribution on a matrix of cost distances between

patches resulting from a combination of riverine distance and barriers (i.e., dams, road crossing structures).

Timesteps for the Brook Trout model were semi-annual and included a spawning period and an overwintering period. Each year prior to spawning, we simulated local dispersal movement followed by size- and density-dependent mortality and – for those years scheduled for electrofishing suppression – density-independent mortality. Spawning then consisted of length-based maturation, mate selection, length-based egg production, and sex chromosome inheritance. Following spawning, individual fish were grown according to a von Bertalanffy growth curve (von Bertalanffy 1938) adjusted for local patch temperature. During the overwintering period, individuals again experienced density-dependent mortality and individual growth. In years scheduled for YY male releases, releases occurred just prior to spawning. All simulations were run for 120 years, with a burn-in phase of 20 years to stabilize population demographics. Suppression treatment began at year 20 and continued for 100 years according to the designated suppression schedule. Releases of YY males began at year 23 and continued annually through the end of the simulation.

Study Design

We conducted simulation experiments to test for effects of alternative management strategies on Brook Trout eradication outcomes. Experiments involved varying the management factors of interest while holding all other model parameters constant. Target management variables included the number and body size of YY males released, frequency and efficacy of electrofishing mortality, number of locations from which YY males were released, and proportion of YY males mature at release (Table 1). We selected parameter ranges after careful consideration of what we considered to be logistically feasible based on expert opinion from local fisheries managers, including co-author WPB. The primary indicators used to measure success of the TYC program were the number of years of treatment required

for Brook Trout eradication or – in the case where eradication did not occur – the minimum number of females observed during the simulation.

For reference, we refer to a ‘standard’ model scenario in which the level of each experimental factor reflects the current management strategy (Baker et al. 2020), which calls for approximately 4,000 YY fingerlings to be released annually in the headwaters of Sullivan Creek (defined as the reach upstream from the confluence with Gypsy Creek; see Figure 1). Since these annual releases represent a limited pilot program, we simulated a more comprehensive treatment that reflects the current management strategy but expanded YY male stocking activities to the entire watershed. Accordingly, we proportionally increased the ‘standard’ number of YY fingerlings released to 9,110, or approximately 80% of the estimated age-1+ abundance and 60% of the estimated age-0 abundance for wild Brook Trout (Schill et al. 2017). The ‘standard’ suppression schedule consisted of five-year suppression cycles with three years of three-pass electrofishing suppression followed by two years off. All YY males were reproductively immature at release. Because the immediate dispersal distribution of YY males is unknown, the standard model assumed they would distribute themselves throughout the watershed in proportion to the known heterogeneous local Brook Trout density across patches. Because the fitness of YY males relative to wild Brook Trout is unknown, we assumed equal fitness in terms of both survival and reproduction (but see Day et al. (2020) for a full treatment of fitness effects).

YY male release number and body size: To test for effects of adjusting the number and body size of YY males released, we varied these two factors simultaneously in a fully crossed experimental design. Body size (mean $\pm$ SD) of YY males comprised two levels: fingerling (137 ± 14 mm) and catchable (230 ± 15 mm), representative of the likely size of fish following one and two years of growth in the hatchery, respectively (WPB *pers. obs.*). Number of fish released was limited by maximum plausible estimates provided by local managers, ranging from 20 – 200% of the size of the standard simulated age-1+ population (at increments of 20%) for fingerlings and 10 – 100% for catchables (at increments of 10%).

We conducted 10 replicates of each combination of size and number of fish and calculated 95% confidence intervals for each outcome variable. We fit an exponential decay curve using the package 'nlstools' in R (Baty et al. 2015) and estimated asymptotes for each curve. To understand how stocking large numbers of YY males affects size-based density dependent mortality of wild Brook Trout, we compared age-based survival of wild Brook Trout between standard fingerling and catchable scenarios (i.e., four times as many fingerlings as catchables) for the first five years of YY male stocking. To do this, we used the 'nlme' package in R to run linear mixed models for each age with year as a random effect (Pinheiro et al. 2017).

Suppression schedule and mortality: To test for effects of electrofishing effort and suppression mortality on Brook Trout eradication, we selected five electrofishing schedules (Table 1). We ran 100 simulations for each schedule with the baseline capture probability ranging from 0.01 to 1. During simulations, this capture probability was multiplied by a size-based probability of capture derived from a gear selectivity curve developed using size-based regression analysis on multi-pass depletion electrofishing surveys. For size classes 0 – 7, gear selectivity values were 0.1, 0.45, 0.75, 0.95, 1, 1, 1, and 1, respectively (Day et al. 2018). Thus, a probability of mortality of 1 did not remove all Brook Trout from a given patch. We plotted % mortality against time to eradication and fit the resulting data to an exponential decay curve. Differences among suppression schedules at the empirically derived capture probability (0.85 for Sullivan Creek; Day et al. 2018) were compared using 95% confidence bands.

Release Locations and YY male reproductive maturity: We tested three levels for number of release locations and six levels for maturity of YY males for effects on Brook Trout eradication. In addition to the standard scenario in which YY males were released in all occupied patches, we tested scenarios of 16 and 32 release locations, distributed according to known Brook Trout density hotspots. The 16-location scenario approximated the distribution strategy of the ongoing management plan. For

194 this experiment, we simulated a restricted movement scenario in which the initial dispersal of YY males
195 followed a Pareto distribution.

196 Based on the range of fingerling maturation observed during TYC program development, we
197 varied the proportion of fingerling YY males that were mature upon release, testing levels of 0, 0.25,
198 0.50, 0.75, 0.90, and 1 (Table 1). For each scenario, we conducted 10 replicates and compared Brook
199 Trout eradication outcomes using Analysis of Variance and *post hoc* pairwise t-tests using the Holm-
200 Bonferroni correction (Holm 1979). For access to model input files used for this study, see
201 Supplementary Material 2.

202 Results

203 Under the standard model scenario (9,110 fingerlings released annually, 0% reproductively
204 mature, spatially distributed across all patches relative to Brook Trout density, standard suppression
205 schedule), mean time ($\pm$ SD) to eradication from the beginning of suppression treatment was 27.7 ± 4.2
206 years. During the first 10 years of YY male releases, mean annual YY male survival varied from 0.10 to
207 0.16 (Table 2). The proportion of offspring ($\pm$ SE) spawned by YY males averaged 0.35 ($\pm$ 0.040) after the
208 first year and increased annually by an average of 0.031 to 0.6 ($\pm$ 0.095) by year 10. The proportion of
209 the population that was male ($\pm$ SE) increased from 0.5 ($\pm$ 0.003) to 0.88 ($\pm$ 0.015) over the same time
210 frame. Excluding YY males, proportions of the wild population at year 10 ($\pm$ SE) were 0.79 ($\pm$ 0.021) for XY
211 males and 0.21 ($\pm$ 0.021) for XX females, or a 3.76:1 male to female ratio.

212 YY male release number and body size: For fingerling scenarios, Brook Trout populations were
213 eradicated across all but the lowest value for number of YY males stocked (Figure 2a). Non-linear
214 regression indicated that time to eradication followed a negative exponential curve and reached an
215 asymptote at $26.2 (\pm 2.90 \text{ SE})$ years of treatment, indicating a threshold occurred when number of YY
216 males stocked equaled approximately 80% of the age 1+ population ($27.7 \text{ years} \pm 1.73$). Exceeding this

threshold by releasing additional YY males produced no further reduction in time to eradication. In simulations where catchable YY males were released, Brook Trout populations were not eradicated during 100 years of treatment for all numbers of fish stocked. Minimum number of females observed in these catchable scenarios followed a negative exponential curve with a predicted asymptote near zero (56 ± 34 females; Figure 2b), indicating that eradication may have occurred with further increases of catchable fish beyond the tested range of values in this simulation. Standard fingerling (9,110 YY males) and catchable (2,294 YY males) YY scenarios performed equally well during the first few years of treatment, but the wild population in catchable scenarios stabilized around a minimum female population size of 4,000 individuals (Figure 3).

Survival among wild Brook Trout over the first five years of stocking varied by both age and body size of the YY males (Figure 3). Stocking with fingerling YY males yielded lower survival among wild Brook Trout from ages 2-7 ($p < 0.002$), but greater survival for age-0 wild Brook Trout ($\beta = 0.19$, $p < 0.001$). Survival of wild Brook Trout did not differ between fingerling and catchable scenarios for ages 1, 8, or 9 ($p > 0.05$), although fish older than age 7 made up $< 3\%$ of the population. Survival of YY males also varied according to body size. Mean fingerling YY male survival ($\pm$ SE) over the first 10 years of releases under the standard scenario was $0.12 (\pm 0.0057)$, 2.6 times higher than survival for catchable fish (0.047 ± 0.0038).

Suppression schedule and mortality: Time to eradication for all suppression schedules decayed exponentially across suppression capture probabilities from 0.01 – 1 (Figure 4). For the capture probability estimated from electrofishing suppression in Sullivan Creek (0.85), annual suppression did not differ in length of treatment to achieve eradication (17.5 years, 95% CI 15.8 – 19.4) as a cycle of five years on and two years off (16.4 years, 14.6 – 18.7). Suppression every other year (28.0 years, 26.3 – 31.8) performed equally well as a cycle of three years on and two years off (26.2 years, 24.5 – 28.7). A cycle of two years on and three years off performed worse than all other scenarios (34.8 years, 33.1 –

39.1). In general, lower rates of mortality resulted in greater separation in time to eradication among suppression schedules than the empirically derived capture probability of 0.85 (Figure 4). The difference in treatment time required for eradication between the best- and worst-performing suppression schedules at the empirically derived capture probability was 16.5 years.

Release locations and YY male reproductive maturity: All simulations that utilized 16 or 32 release locations combined with a restricted dispersal scenario for initial YY male distribution failed to eradicate the wild Brook Trout population. Scenarios that incorporated 32 rather than 16 release locations yielded lower population sizes but did not result in eradication ($t_{37} = 6.20$ $p < 0.001$; Figure 5). The proportion of patches occupied by wild Brook Trout over time also decreased with an increase in the number of release locations ($t_{34} = 19.14$, $p < 0.001$; Figure 5). With YY males distributed throughout all patches, eradication of wild Brook Trout occurred after an average of 27.7 (± 4.2 SD) years of treatment.

The proportion of fingerling YY males that were mature upon release was inversely correlated with the number of years of YY stocking required for eradication ($F_{1,58} = 199.3$, $p < 0.001$). Significant decreases in time to eradication occurred from 0 to 0.25, 0.25 to 0.5, 0.5 to 0.9, and 0.5 to 1 mature fingerlings ($p < 0.05$; Figure 6). Increasing this proportion from 0 to 0.5 decreased stocking time by 37.6%, from 25.5 years to 15.9 years.

Discussion

Modeling the release of YY males in tandem with mechanical suppression to achieve eradication of non-native Brook Trout showcases a wide range of potential outcomes dependent upon management decisions. Our results indicate that these decisions can significantly impact the duration and effectiveness of a TYC program. Furthermore, they make it clear that understanding the key factors and underlying mechanisms via simulation modeling is essential for adaptively managing a TYC program.

While simulation models such as ours can be used to make predictions, they are often more useful in evaluating the dynamics of complex systems and the range of outcomes and uncertainty associated with different management strategies (Melis et al. 2015; Starfield 1997; Walters 1986).

Number of YY males: The number of YY males to release annually is one of the most important questions a TYC program faces. Our simulation results indicated that effectiveness of the program plateaued as the annual number of YY males released increased beyond approximately 80% of the age-1+ population size. This non-linear relationship may have important implications for managers that have to balance time required to achieve eradication with financial cost. If the age-1+ population size can be estimated, the release of YY males can be optimized to allow remaining program resources to be allocated to other aspects of the management plan (Schill et al. 2017).

The mechanism responsible for this pattern – limited returns on additional releases – was size- or stage-specific density-dependent mortality (Ray and Hastings 1996; Stige et al. 2019). In this framework, the number of individuals that could occupy a size class within a given patch was limited by intra-cohort competition. As a result, a threshold for the optimal number of YY fingerlings to release became apparent. Exceeding this threshold failed to improve eradication outcomes due to increased competition, and therefore, increased mortality for the YY male fingerlings. Inter-cohort competition followed the assumptions of the adult dominance model, so that high densities of juvenile YY males had no effect on the mortality of larger size classes. Similar patterns of mortality have been observed in field studies for various salmonids following stocking of hatchery fish (Justice et al. 2009; Kelly-Quinn and Bracken 1989; Miller 1958). In our model, this mechanism caused the number of reproductive YY males in the system to plateau despite simulating larger numbers of fish released.

Size of YY males and competition: Simulations revealed that stocking fingerling YY males was more effective than stocking catchables. The underlying mechanism driving this outcome is the same as that

which regulated the threshold for number of YY males to release: size- and density-dependent mortality. Because population age structure allocates relatively little space to the catchable size class, the large majority of catchable YY males died during their first winter, as is common for larger hatchery fish stocked into streams (High and Meyer 2009; Kennedy et al. 2018). Empirical estimates of YY male Brook Trout survival in the stream environment are currently scarce. In a recent field experiment, Kennedy et al. (2018) observed 4-month survival (June - October) of catchable YY males as high as 0.25, though this result is not directly comparable to our study which reports 12-month survival.

In our simulations, despite the high observed mortality rate, catchable simulations yielded a strongly skewed sex ratio and rapid population decline over the first several years of treatment, due to their ability to spawn before their first winter in the stream. Unlike in the fingerling scenarios, however, the catchable scenarios failed to eradicate the wild Brook Trout population. Instead, the population surprisingly maintained a stable, reduced number of females throughout the simulation (Figure 3a). The reason for this result is that increasing the number of catchable-sized fish beyond carrying capacity put no additional competitive pressure on smaller - but reproductively mature - fish. Day et al. (2020) conducted a substitutive simulation experiment to show that mortality due to competition and replacement of the wild population with YY males – and not just sex ratio manipulation – makes a significant contribution to the decline of the wild population. In our size-based model of competition, early maturing Brook Trout (ages 2 - 4) experienced additive mortality due to intra-cohort competition in the fingerling YY scenarios, but not in the catchable YY scenarios (Figure 3b). This additional source of mortality on younger fish was enough to drive the entire population to extinction. While our model does implement inter-cohort competition under a normal size structure scenario, the negative correlation between density of larger size classes and survival of smaller size classes is not maintained when the larger classes exceed carrying capacity (Nordwall et al. 2001).

Our results raise important questions about competition among YY male and wild fish, and therefore, questions about the size of fish to release relative to the size of fish in the wild population. Standard models of salmonid mortality assume that density-dependence is important only in juvenile stages, and that inter-cohort competition is not a significant factor (Elliott 1985; Milner et al. 2003). However, some experimental and field-based studies have observed effects of inter-cohort competition on individual growth (Kaspersson et al. 2013; Myrvold and Kennedy 2015), habitat selection (Ayllon et al. 2013; Hojesjo et al. 2016), and survival (Nordwall et al. 2001; Samhoury et al. 2009) as a result of either adult (Sandlund et al. 1991; Kaspersson et al. 2013) or juvenile dominance (Einum et al. 2011; Kvingedal and Einum 2011). These studies usually involve adjacent cohorts rather than interactions between large and small adults (Housego 2019). However, at least one study has found that inter-cohort competition between 150 mm and 230 mm roach (*Rutilus rutilus*) manifested in reduced individual growth rates for both size classes (Hjelm and Persson 2001). Whether inter-cohort competition occurs – and in which direction – is ultimately dependent upon a host of factors including species, habitat and resource availability, ontogenetic niche shifts in prey and habitat preference, and reproductive strategy (Sandlund et al. 1991; Einum and Kvingedal 2011). While our model’s assumption that an influx of large, YY males has no competitive effect on smaller adults is likely unrealistic, it underlines the need for a greater understanding of the effects of density-dependence on eradication of the wild population. Further simulation work and field studies are needed to test alternative hypotheses of inter-cohort competition between hatchery and wild fish, particularly to inform eradication programs.

Suppression schedule: Another factor that managers should consider is the design of electrofishing suppression that is coupled with YY releases. Previous modeling work has shown that combining YY releases with mechanical removal has a positive synergistic effect on eradication outcomes (Day et al. 2020; Schill et al. 2017). Our results demonstrate that time to eradication is highly dependent on the level of mortality caused by suppression. In addition, the frequency with which suppression is applied

can affect time to eradication, with annual suppression reducing treatment time by 16.5 years in comparison to a 5-year cycle of 2 years of suppression followed by 3 years off. Frequent suppression in consecutive years may therefore be an attractive option, but managers should also consider the potential for unintended consequences. In particular, native fishes could be negatively impacted by frequent electrofishing suppression through injury and mortality (Keefe et al. 2000; Kocovsky et al. 1997) as well as undergo behavioral and physiological changes (Dwyer et al. 2001; Mesa and Schreck 1989). We found that a cycle of 5 years on and 2 years off performed as well as an annual suppression program, which would reduce the years requiring suppression effort by 29% (e.g., 8 years of effort saved over a 30-year treatment program).

Reproductive maturity of YY males: Reproductive maturity of YY fish is another factor likely to affect the amount of time and effort required to eradicate non-native species. Our standard model assumes that no YY male fingerlings would be mature upon release. Reports from hatcheries rearing YY males have indicated, however, that some proportion of fingerlings appear to be mature (expressing milt) at sizes equivalent to the fingerlings simulated in this study, and that perhaps the proportion of mature fingerlings could be manipulated (e.g., through food ration). In our model, increasing the proportion of mature fingerlings from 0 to 0.5 decreased time to eradication by 38% to 16 years of stocking. This result is a promising sign that early maturation could reduce the overall effort required to achieve program goals by producing more YY spawners each year while also increasing density-dependent mortality on young, mature wild Brook Trout. However, reproductive fitness of mature YY fingerlings could play an important role here as well (Day et al. 2020).

Number of release locations: Finally, we hypothesized that increasing the number of release locations would contribute to a more rapid decline of Brook Trout. While spreading YY releases evenly throughout the watershed was most effective, doubling the number of release locations from 16 to 32 reduced the mean number of females in the population and the number of occupied patches. However, both

scenarios failed to achieve eradication. This result is likely due to a combination of the high mortality experienced by YY males when released in high densities to a few locations and the restricted dispersal distribution we used to regulate movement of individuals prior to their first spawning season. Earlier simulations have shown that greater YY mobility leads to faster declines in the wild Brook Trout population (Day et al. 2020), but hatchery fish are likely to be relatively sedentary (Helfrich and Kendall 1982; Symons 1969). Early monitoring of YY males in Sullivan Creek has not measured dispersal prior to spawning, but has indicated that YY movement over a 9-10-month period has been limited with very little long-distance dispersal (Baker et al. 2020; Lukacs 2020). We recommend that for maximum efficiency, managers disperse releases of YY males as widely as possible and relative to local densities of the non-native species.

Model development and management implications: While our modeling results present many alternative projections (e.g., years required for eradication), we emphasize that the main purposes of the model are to explore system dynamics and to provide management recommendations based on a comparison of relative outcomes across a range of scenarios (Starfield 1997; Walters 1986). Field studies and monitoring programs for a variety of habitats and species are needed to test model assumptions and thereby improve model projections. In Sullivan Creek, monitoring work to support adaptive management is ongoing, and has provided preliminary data on YY male survival that aligns well with the survival rates that emerged from model processes (Lukacs 2020). This result provides support for the size- and density-dependent mortality mechanism that produced higher mortality for YY fish than wild fish.

The assumption of equal fitness between hatchery and wild fish also carries uncertainty and could be informed by the monitoring of released YY males through time. Hatchery fish tend to have lower fitness than wild fish (Araki et al. 2008), but one study did report YY males having greater reproductive fitness than their XY counterparts (Hamilton et al. 1969). Because effects of YY male fitness

on Brook Trout eradication were treated in an earlier application of the model (Day et al. 2020), we did not explore this factor further, but we highlight its potential to impact a TYC program.

Programs to remove non-native species such as the program simulated here must also consider their potential impacts on the native species they are intended to benefit. In many systems across their native range including our study system, Westslope Cutthroat Trout compete with non-native Brook Trout for space and resources (Dunham et al. 2002). The introduction of a large population of YY male Brook Trout would be expected to increase competition pressure with the native Westslope Cutthroat Trout. In addition, the effects of repeated electrofishing to suppress the wild Brook Trout population could negatively impact the Westslope Cutthroat Trout due to unintended electrofishing-related injuries or mortality (Kocovsky et al. 1997). Programs to monitor the status of native populations should be employed in conjunction with a TYC program to address these potential impacts. Doing so will aid managers in finding a balance between eradicating the non-native species and preserving the demographic and genetic integrity of the native species. In our study system, the native Westslope Cutthroat Trout are monitored annually to guide the suppression and TYC management plans (Baker et al. 2020).

In an adaptive management framework, model results and ongoing management activities work mutually to reduce uncertainty and improve decision-making. Ongoing Brook Trout model development includes an application to simulate multiple competitor species and their interactions during TYC implementation. We are further exploring how – in the case of failure to eradicate – Brook Trout recovery might be affected by reduced fitness following a population bottleneck due to suppression (also see Day et al. 2018). We encourage managers to consider incorporating simulation modeling into TYC and similar programs to improve understanding of underlying system processes on potential management decisions. Our study shows that decisions affecting just a single factor (e.g., the number to

release, release locations, or the proportion of reproductively mature individuals) can significantly reduce the time required to achieve eradication, or result in failure to achieve eradication.

While our study focuses on removal of Brook Trout, we anticipate that sex ratio distortion programs will be used for the local eradication of a wide variety of undesirable species (Harvey-Samuel et al. 2017; Galizi et al. 2016). Aquatic systems are especially well-suited to the TYC strategy because of their linear nature which forces interaction among individuals as a result of limited space and resources. Indeed, programs have previously been explored for amphibians (Thresher and Bax 2006) and several other species of fish (Thresher et al. 2019). TYC programs have potential to become a valuable tool for conservation by reducing the financial cost and ecological impact of non-native species eradication, but success will depend in part on the depth to which underlying system mechanisms are understood, particularly in response to the TYC approach. Results from this study can inform future TYC programs on identifying some of the key factors to consider when making decisions, especially where the availability of local empirical data may be limited.

Acknowledgments

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600 **Tables**

601 **Table 1.** Factors tested within the CDMetaPOP modeling framework for their effect on Brook Trout
 602 eradication using mechanical suppression and stocking of YY males.

Factor	Levels	Description
No. of YY males released annually	10 increments from 10 – 200% of wild age-1 population size	Percentages for fingerling and catchable YY males ranged from 20-200 and 10-100, respectively
Body size of YY males released annually	137 ± 14 mm (fingerling)* 230 ± 15 mm (catchable)	Sizes were based on estimates from 1 (age-0) and 2 (age-1+) years of hatchery growth
Suppression cycle schedule	2/3, 3/2*, 1/1, 5/2, annual	Values represent no. of years of suppression followed by no. of years without suppression
Suppression mortality	100 increments from 1 – 100%	Mortality value was multiplied by a size-based gear selectivity curve
No. of stocking locations	All patches*, 16, 32	Stocking locations were selected according to known Brook Trout density hotspots
Maturity of YY males	0*, 0.25, 0.50, 0.75, 0.90, 1	Proportion of released YY males that were reproductively mature

603 * indicates standard suppression program

604

Table 2. Mean output metrics ($\pm$ SE) resulting from 10 replicates of the standard model scenario of EBT suppression and YY male releases throughout the Sullivan Creek Watershed. Only the first 10 years of releases are presented. An X indicates whether a given treatment occurred in a given year.

Treatment Year	Suppression*	YY	Proportion male (excluding YY)	Proportion male†	Proportion YY progeny	YY annual survival
1	X		0.50 (0.003)	0.50 (0.003)	0	NA
2	X		0.50 (0.004)	0.50 (0.004)	0	NA
3	X		0.50 (0.005)	0.50 (0.005)	0	NA
4	X	X	0.50 (0.005)	0.53 (0.005)	0	NA
5	X	X	0.56 (0.006)	0.60 (0.005)	0	0.12 (0.004)
6	X	X	0.66 (0.011)	0.69 (0.011)	0.35 (0.040)	0.16 (0.005)
7	X	X	0.70 (0.013)	0.73 (0.013)	0.45 (0.045)	0.10 (0.005)
8	X	X	0.71 (0.015)	0.74 (0.014)	0.40 (0.028)	0.09 (0.006)
9	X	X	0.71 (0.012)	0.76 (0.011)	0.39 (0.034)	0.10 (0.007)
10	X	X	0.71 (0.010)	0.78 (0.007)	0.42 (0.049)	0.13 (0.005)
11	X	X	0.74 (0.010)	0.80 (0.010)	0.46 (0.043)	0.15 (0.006)
12	X	X	0.77 (0.021)	0.82 (0.015)	0.54 (0.047)	0.13 (0.007)
13	X	X	0.77 (0.020)	0.85 (0.012)	0.58 (0.074)	0.12 (0.003)
14	X	X	0.79 (0.021)	0.88 (0.015)	0.60 (0.095)	0.13 (0.004)

* Suppression was on a 3/2 year on/off cycle, but was offset among subwatersheds, resulting in suppression occurring every year.

† This proportion includes YY males, but only those that survive over winter.

Figure captions

Figure 1: Simulation extent and stocking locations for the Brook Trout TYC model in the Sullivan Creek Basin in northeastern Washington State, USA. Light grey shading along Sullivan Creek represents the treatment extent and the current range of Brook Trout in Sullivan Creek. Triangles indicate release locations for YY male Brook Trout for simulations that tested for effects of the number of release locations used. This map was created using ArcGIS Desktop version 10 (Redlands, CA).

Figure 2: Simulation results from a YY male Brook Trout suppression program in the Sullivan Creek Basin in Washington State. The x-axis represents the number of YY males released relative to the size of the baseline number of age-1+ fish in the population. The top panel (a) displays the amount of time required to achieve eradication using fingerling-sized fish (137 ± 14 mm), with an asymptote occurring at 26.2 years. The bottom panel (b) displays results when using catchable-sized age-1 fish (average 230 ± 15 mm), but presents the minimum number of observed females since these simulations did not result in eradication. Error bars indicate 95% confidence intervals derived from 10 Monte Carlo replicates. For reference, the simulated baseline number of age-1+ Brook Trout in Sullivan Creek is approximately 11,000. Average number of females in the population prior to treatment is approximately 12,800.

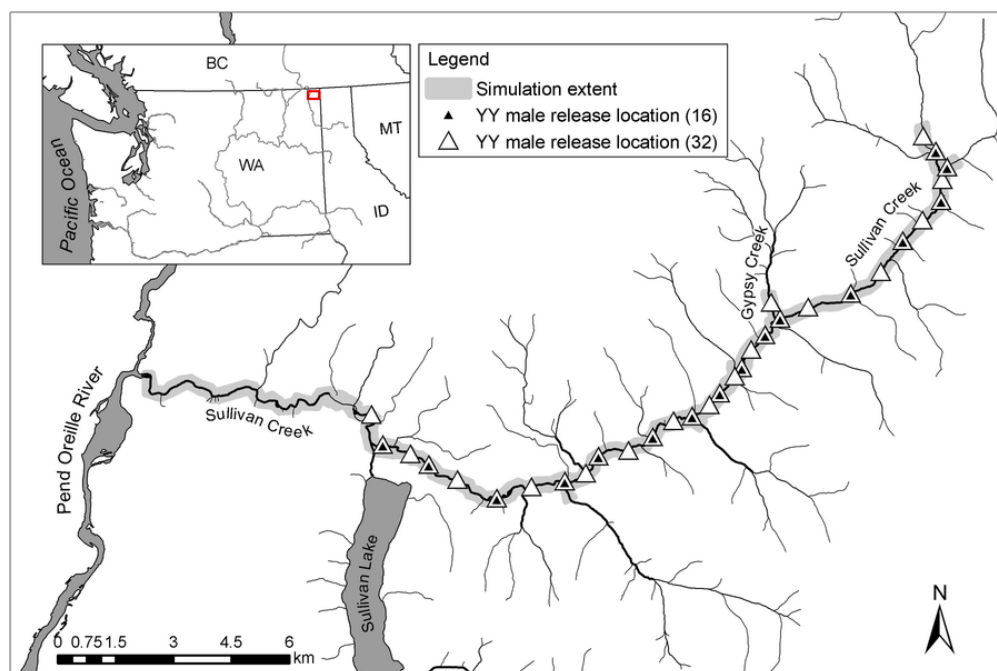
Figure 3: Comparison of simulations using mechanical removal plus YY male fish to eradicate non-native Brook Trout in the Sullivan Creek Basin in Washington State. Suppression treatment began at year 20; releases of YY males began at year 23. Four times as many YY males were released in the fingerling scenario (body size = 137 ± 14 mm) as in the catchable scenario (230 ± 15 mm). The top panel shows population size over time, while the bottom panel shows age-specific survival during the first five years of stocking. Catchable YY males were mature upon release and therefore induced a stronger immediate effect on sex ratio than fingerlings. Catchable scenarios did not completely eradicate Brook Trout, however, because they did not increase density-dependent mortality upon young, mature wild Brook

636 Trout as occurred with the stocking of fingerling YY males. Differences in survival between scenarios
637 were due primarily to density-dependence resulting from competition with YY males. Results shown are
638 averages across 10 Monte Carlo replicates.

639 Figure 4: Effect of suppression schedule on time to eradication of a simulated Brook Trout population in
640 the Sullivan Creek Basin in Washington State under a range of suppression mortality scenarios. The gray
641 vertical line indicates the empirically derived capture probability (0.85). Solid lines represent exponential
642 decay curves fit to the output data, while dotted lines indicate 95% confidence bands.

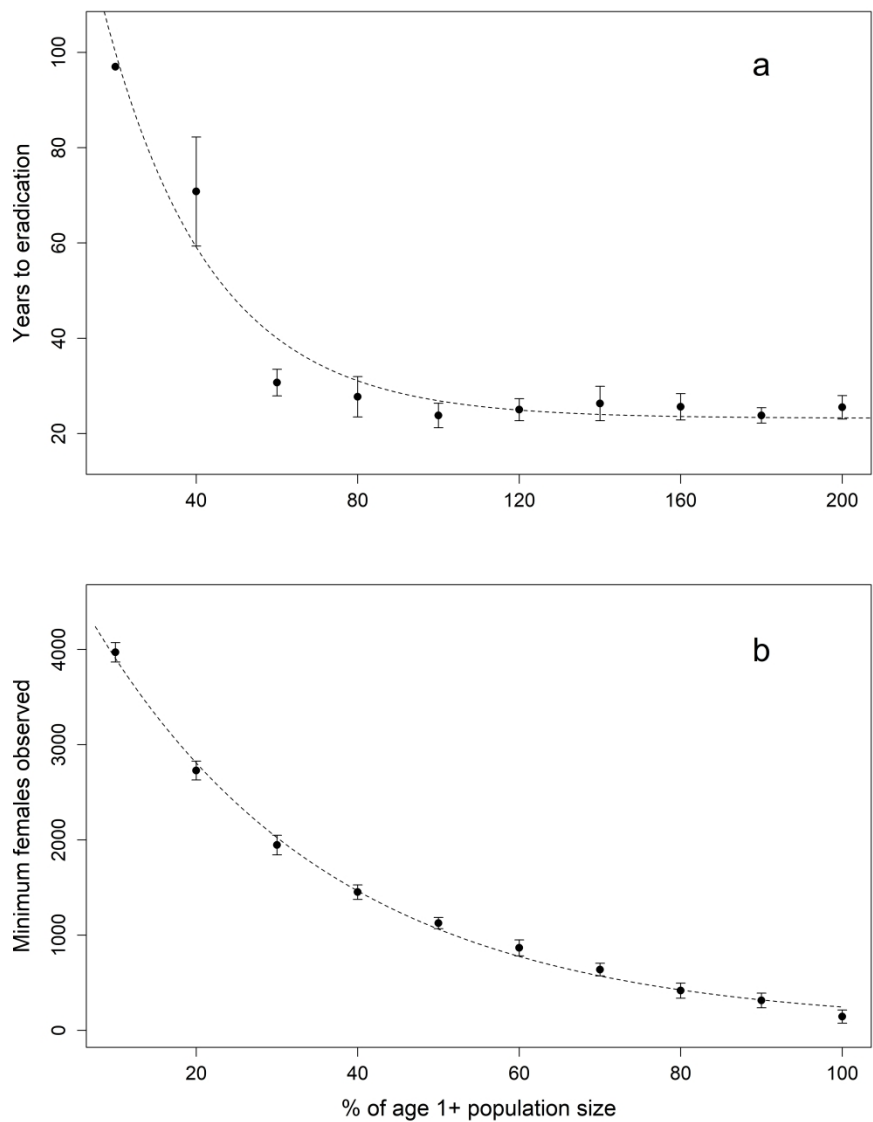
643 Figure 5: Effect of varying the number of locations from which YY male Brook Trout were released in the
644 Sullivan Creek Basin in Washington State. Solid lines represent mean values derived from 10 Monte
645 Carlo replicates, and polygons indicate 95% confidence bands. Release locations were selected based on
646 proximity to one another and on known Brook Trout density. The top panel omits the final half of the
647 simulation to provide greater resolution and because the populations maintain a steady state after year
648 60. Suppression treatment began at year 20; releases of YY males began at year 23.

649 Figure 6: Simulation results ($\pm$ SD) indicating the effect of the proportion of YY fingerlings that are
650 reproductively mature at release on the amount of treatment time required to eradicate a population of
651 Brook Trout in the Sullivan Creek Basin in Washington State. Suppression treatment began at year 20;
652 releases of YY males began at year 23. Overall, adding 50% mature fingerlings reduced time to
653 eradication by an average of 37%. The bottom panel provides additional detail on the number of
654 individuals containing each chromosomal arrangement over time.

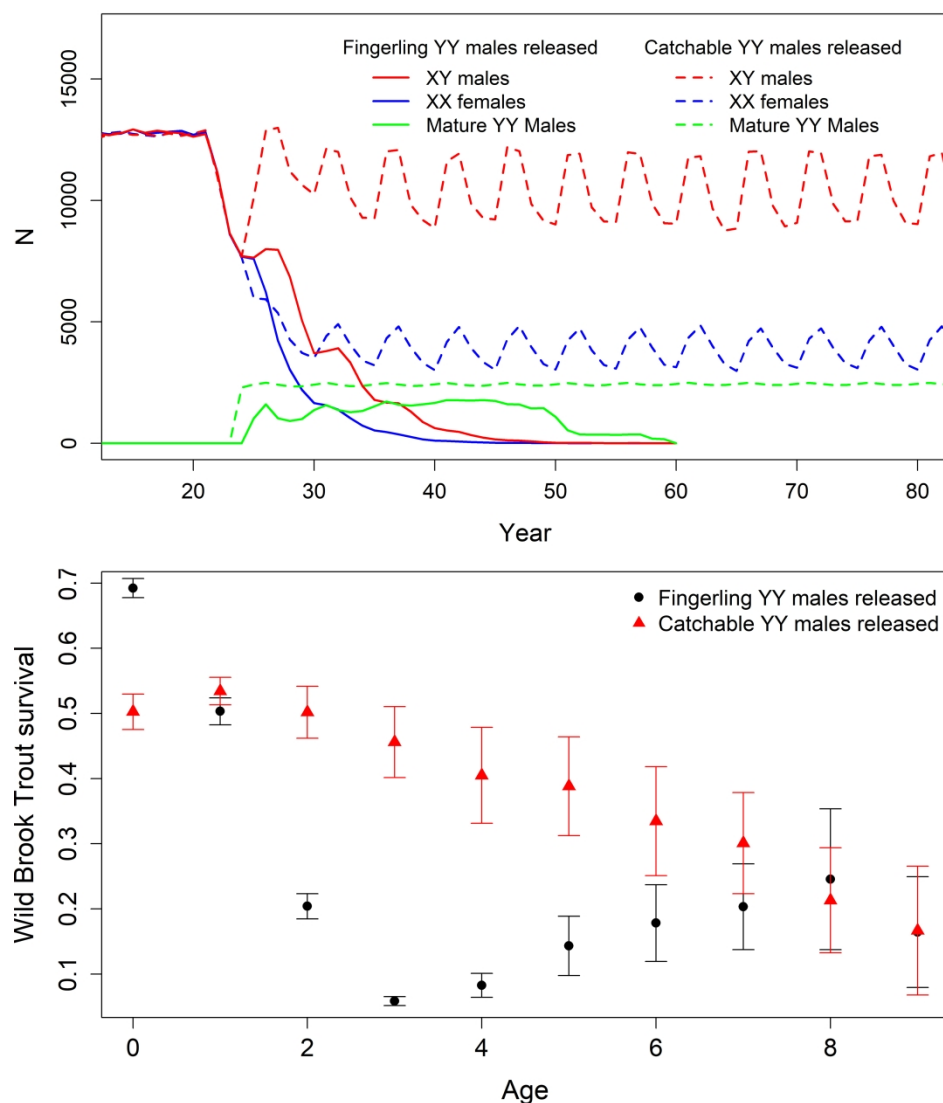


Simulation extent and stocking locations for the Brook Trout TYC model in the Sullivan Creek Basin in northeastern Washington State, USA. Light grey shading along Sullivan Creek represents the treatment extent and the current range of Brook Trout in Sullivan Creek. Triangles indicate release locations for YY male Brook Trout for simulations that tested for effects of the number of release locations used. This map was created using ArcGIS Desktop version 10.6.

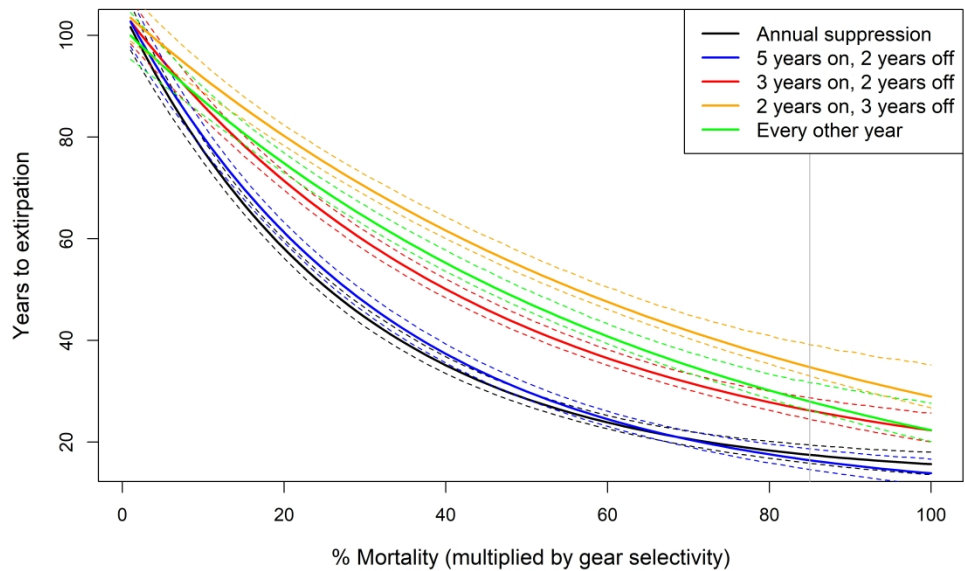
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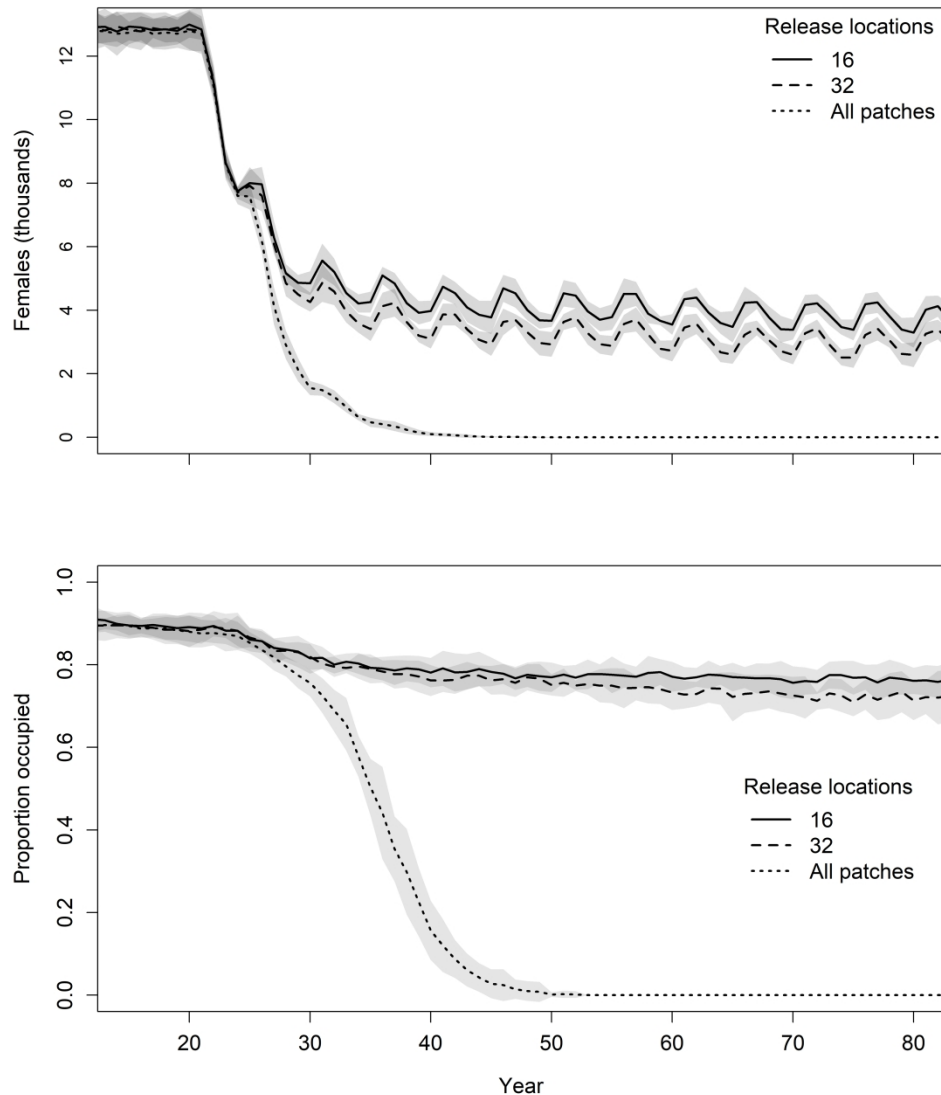
Simulation results from a YY male Brook Trout suppression program in the Sullivan Creek Basin in Washington State. The x-axis represents the number of YY males released relative to the size of the baseline number of age-1+ fish in the population. The top panel (a) displays the amount of time required to achieve eradication using fingerling-sized fish (137 ± 14 mm), with an asymptote occurring at 26.2 years. The bottom panel (b) displays results when using catchable-sized age-1 fish (average 230 ± 15 mm), but presents the minimum number of observed females since these simulations did not result in eradication. Error bars indicate 95% confidence intervals derived from 10 Monte Carlo replicates. For reference, the simulated baseline number of age-1+ Brook Trout in Sullivan Creek is approximately 11,000. Average number of females in the population prior to treatment is approximately 12,800.



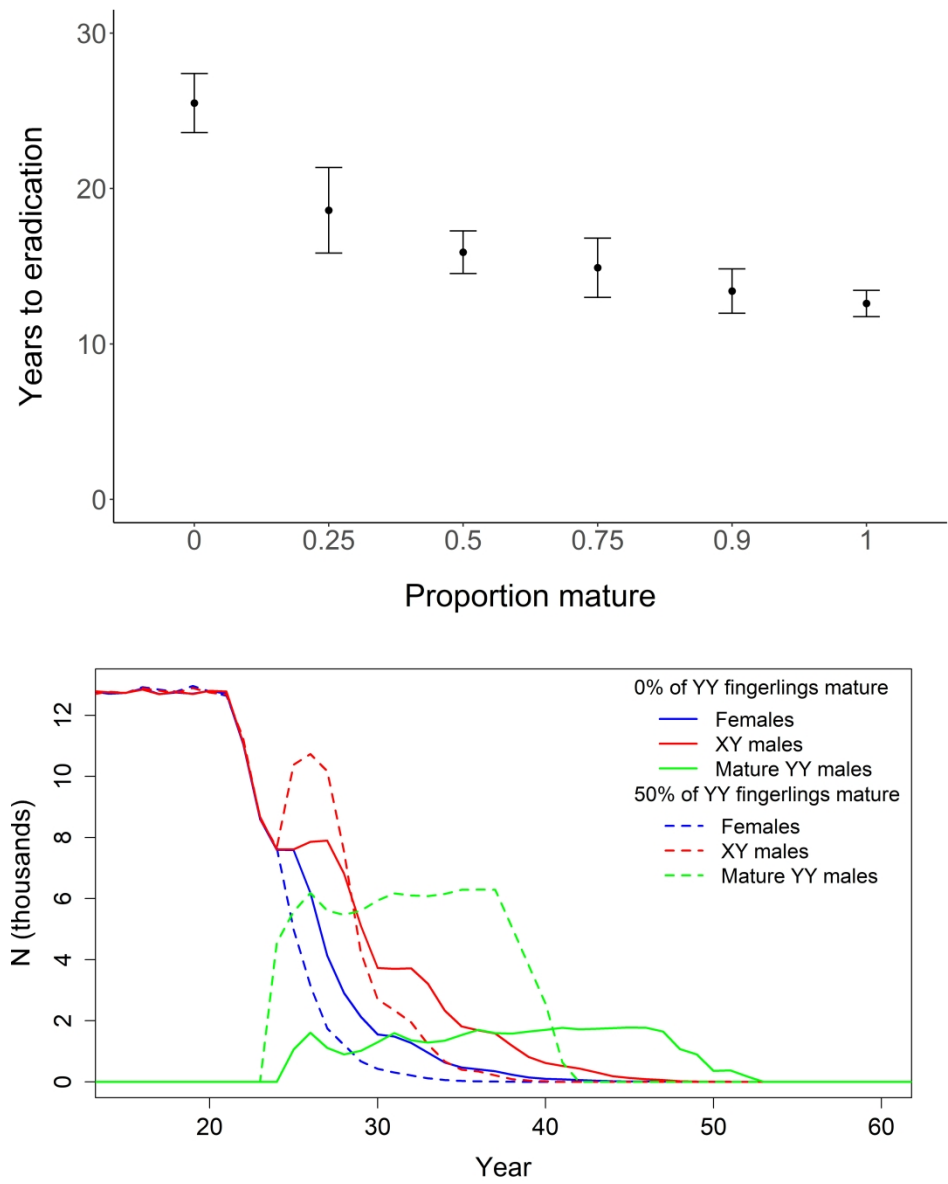
Comparison of simulations using mechanical removal plus YY male fish to eradicate non-native Brook Trout in the Sullivan Creek Basin in Washington State. Suppression treatment began at year 20; releases of YY males began at year 23. Four times as many YY males were released in the fingerling scenario (body size = 137 ± 14 mm) as in the catchable scenario (230 ± 15 mm). The top panel shows population size over time, while the bottom panel shows age-specific survival during the first five years of stocking. Catchable YY males were mature upon release and therefore induced a stronger immediate effect on sex ratio than fingerlings. Catchable scenarios did not completely eradicate Brook Trout, however, because they did not increase density-dependent mortality upon young, mature wild Brook Trout as occurred with the stocking of fingerling YY males. Differences in survival between scenarios were due primarily to density-dependence resulting from competition with YY males. Results shown are averages across 10 Monte Carlo replicates.



Effect of suppression schedule on time to eradication of a simulated Brook Trout population in the Sullivan Creek Basin in Washington State under a range of suppression mortality scenarios. The gray vertical line indicates the empirically derived capture probability (0.85). Solid lines represent exponential decay curves fit to the output data, while dotted lines indicate 95% confidence bands.



Effect of varying the number of locations from which YY male Brook Trout were released in the Sullivan Creek Basin in Washington State. Solid lines represent mean values derived from 10 Monte Carlo replicates, and polygons indicate 95% confidence bands. Release locations were selected based on proximity to one another and on known Brook Trout density. The top panel omits the final half of the simulation to provide greater resolution and because the populations maintain a steady state after year 60. Suppression treatment began at year 20; releases of YY males began at year 23.



Simulation results ($\pm$ SD) indicating the effect of the proportion of YY fingerlings that are reproductively mature at release on the amount of treatment time required to eradicate a population of Brook Trout in the Sullivan Creek Basin in Washington State. Suppression treatment began at year 20; releases of YY males began at year 23. Overall, adding 50% mature fingerlings reduced time to eradication by an average of 37%. The bottom panel provides additional detail on the number of individuals containing each chromosomal arrangement over time.